

Maaz Saeed

MACHINE LEARNING ENGINEER

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📍 Ilmenau, Germany - Open to relocate

EDUCATION

MSc Research in Media Engineering

Technical University Ilmenau, Germany
Apr 2023 – Present

BS Information Technology

International Islamic University, Islamabad
Sep 2017 – Aug 2021

TECHNICAL SKILLS

Machine Learning

PyTorch · Transformers · CNNs · Scikit-Learn

Languages

Python · C++ (Basic)

Audio Processing

Librosa · SciPy · NumPy · Matplotlib

Computer Vision

OpenCV · Image processing · Object detection · Open3D · Image segmentation · Deep SORT

Other

Django · Flask · React · Docker · Git · ONNX · Streamlit

SOFT SKILLS

- Cross-domain adaptability
- Experimental rigour
- Reproducibility-minded
- Independent research

LANGUAGES

English C1
German A1

PROFILE

Machine learning engineer at Fraunhofer IDMT. Consistently stepped into new problem domains — speech and image denoising, sound event and object detection, camera–LiDAR 3D tracking — and delivered measurable gains. Now targeting data scarcity with few-shot sound event detection.

EXPERIENCE & RESEARCH

Master's Thesis Student

Jul 2026 – Present

Fraunhofer IDMT — Few-Shot Class-Incremental Learning for Sound Event Detection

- Adapting few-shot class-incremental learning from vision to sound event detection, an underexplored setting
- Designing base/novel class splits and 1–5-shot incremental sessions on DESED and ESC-Soundscapes
- Adding metrics for forgetting, accuracy retention, and base/novel-class interference across sessions

Working Student – Machine Learning

Jan 2026 – Jun 2026

Fraunhofer IDMT — Sound Event Detection & Dataset Construction

- Fine-tuned YOLOv11 and RT-DETR on log-Mel spectrograms; benchmarked against BEATs and ATST baselines
- Designed post-processing (NMS tuning, aspect-ratio filtering, box merging): event F1 0.517 → 0.679
- Built annotated detection datasets from three datasets; defined project-wide labelling conventions
- Ran multi-GPU ablations in reproducible PyTorch pipelines; first-authored BioDCASE 2026 technical report

Research Project – Machine Learning

Jun 2025 – Dec 2025

Fraunhofer IDMT — Multi-Task Detection & Classification

- Built end-to-end pipeline for joint 26-class tagging and foreground/background prediction on USM-SED dataset
- Raised single-task SET mAP 0.276 → 0.384 via GAP+GMP pooling and SpecAugment/grid-distortion augmentation
- Dual-head multi-task training lifted SET mAP 0.398 → 0.410 and FGBG 0.351 → 0.371 on an identical split
- Validated with 6-fold CV (SET 0.416 ± 0.005 mAP), beating the published USM-SED baseline of 0.37

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PUBLICATIONS

BioDCASE 2026

Task 2 Technical Report

Comparison of Vision-Based
Object Detectors and an Audio
Transformer for Whale Sound
Event Detection — 2026

Authors: Maaz Saeed, Amir Latifi
Bidarouni, Hanna Lukashevich,
Jakob Abeßer

DCASE 2026 Workshop

Extended manuscript under
review

HOBBIES

Travel & the outdoors

Exploring cities and landscapes
on foot, hiking in the Thuringian
Forest and elsewhere.

Competitive Tekken 8

Competed in the Esports World
Cup Last Chance Qualifier, Paris
2026.

HACKATHONS

Kickelhack 2024, Ilmenau

HackCarpathia 2026, Rzeszów

EARLIER EXPERIENCE

Full-Stack Web Developer – Freelance

Sep 2021 – Feb 2023

- Built and maintained client websites and web applications (React, Django, REST APIs).
- Developed and maintained web platforms across multiple domains, including a School Management System for HOPE School and e-commerce websites.

ACADEMIC PROJECTS

Camera–LiDAR 3D Tracking

Apr 2025 – Oct 2025

- Built an end-to-end KITTI pipeline: YOLO 2D detection, LiDAR depth association, class-aware Deep SORT tracking.
- Implemented LiDAR-to-camera projection from raw KITTI calibration and rectification matrices, not prebuilt libraries.
- Benchmarked YOLO11s vs YOLO26s on a fixed split: Car MOTA 28.37 → 33.95, IDF1 54.46 → 58.80, 7.0% faster.
- Detector swap alone cut false positives 74 → 60 and identity switches 13 → 7 at identical tracker settings.
- Shipped a TrackEval harness with 129 tests on GitHub Actions CI, plus a Streamlit model-comparison dashboard.

U-Net Denoising Audio and Image

Oct 2024 – Feb 2025

- Trained U-Net encoder–decoders for two denoising tasks: speech enhancement and Gaussian image denoising
- Built paired training data from LibriSpeech mixed with ESC-50 noise at varied SNRs, and calibrated synthetic image noise
- Converted audio to fixed-size STFT spectrograms, letting one convolutional architecture serve both domains
- Used residual skip connections to preserve fine detail and stabilise gradients through the decoder

HACKATHON PROJECTS

Bloom – Relationship Reminder PWA

HackCarpathia 2026 – 24-hour

Rzeszów, Poland

- Shipped a full-stack installable PWA in 24 h: Django REST backend, React frontend, PostgreSQL in Docker
- Built token auth, 19 REST endpoints, seeded demo data and a service worker for offline install
- Designed the core mechanic – contact cadence, streak scoring and decay – rendered as a growing plant garden
- Scoped for ADHD-friendly use: fast capture, gentle nudges, low-shame feedback over punitive streaks